Programming in the Era of AI

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Normal vs. revolutionary science

Thomas Kuhn, *The Structure of Scientific Revolutions* (1962)

- **Normal science**: puzzle-solving inside an accepted paradigm. Shared tools, shared methods, shared idea of what counts as a good answer.
- **Revolutionary science**: anomalies pile up, the old paradigm cracks, a new one replaces it.
- After a revolution, the old experts are not always the best guides.

Is programming in a period of normal science, or of revolution?

Where I am coming from

- Author of high-performance libraries integrated into major browsers, runtimes and standard libraries: **simdutf**, **fast_float**, **simdjson**, **Roaring Bitmaps**.
- Among the top 2% of scientists worldwide (Stanford/Elsevier), 100+ peer-reviewed papers.
- Among the 1000 most-followed developers on GitHub.
- **And: I have not typed most of my code by hand in over a year.**

Preface

Back in the 1990s

The singularity

> *« I believe that the creation of greater than human intelligence will occur during the next thirty years. (I'll be surprised if this event occurs before 2005 or after 2030.) »*
>
> Vernor Vinge, 1993

Part 1

Where we actually are



Daniel Lemire
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Boost

What fraction of your code is typed by hand (no AI)



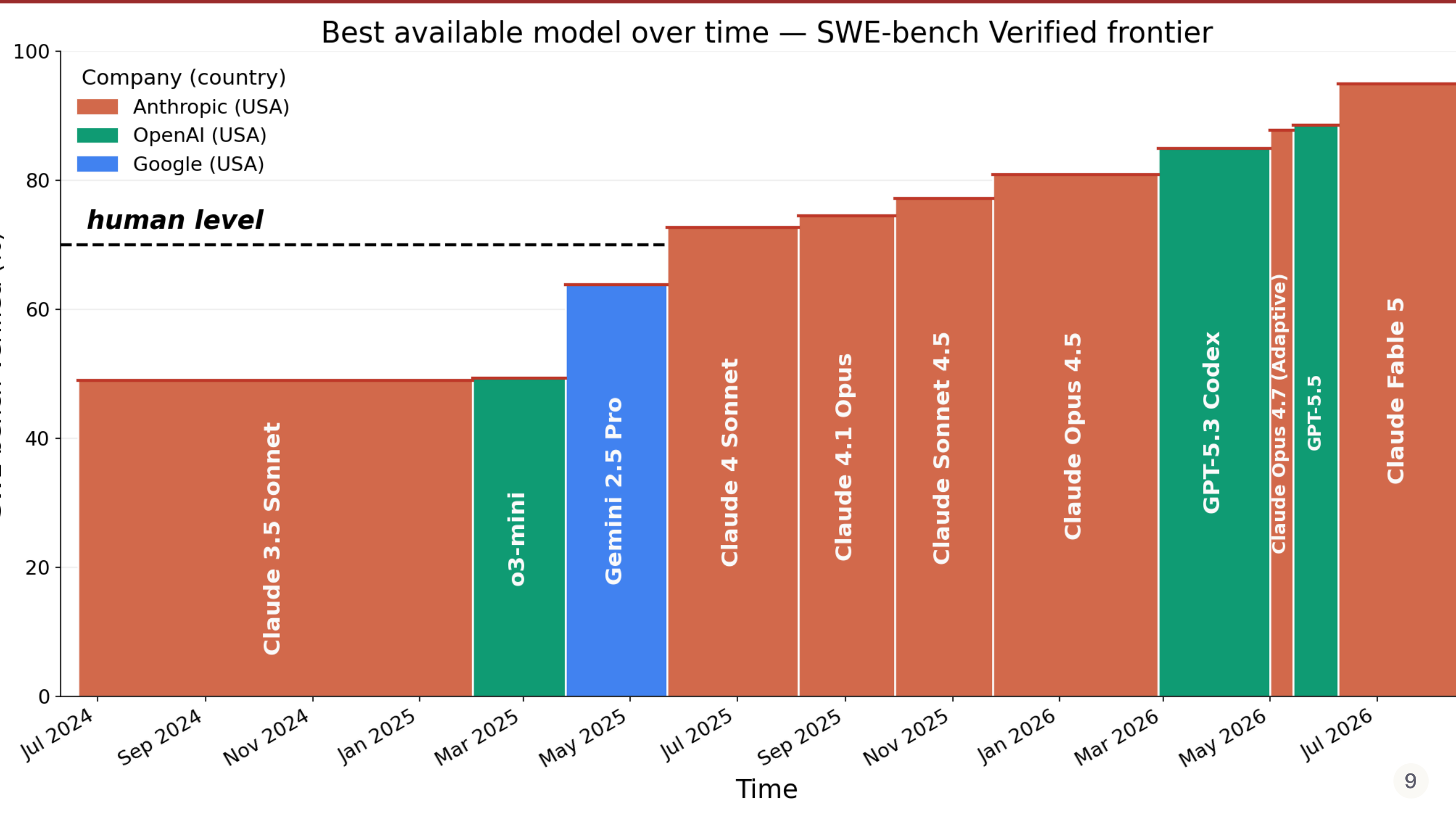
484 votes · Final results

8:59 PM · Oct 19, 2026 · **3,809** Views

16 11 6

SWE-bench Verified

- Real GitHub issues from real open-source Python repositories.
- The model gets the repository, the issue text, and nothing else.
- Success = the project's own **hidden tests** pass afterwards.
- 500 tasks, human-validated as solvable.

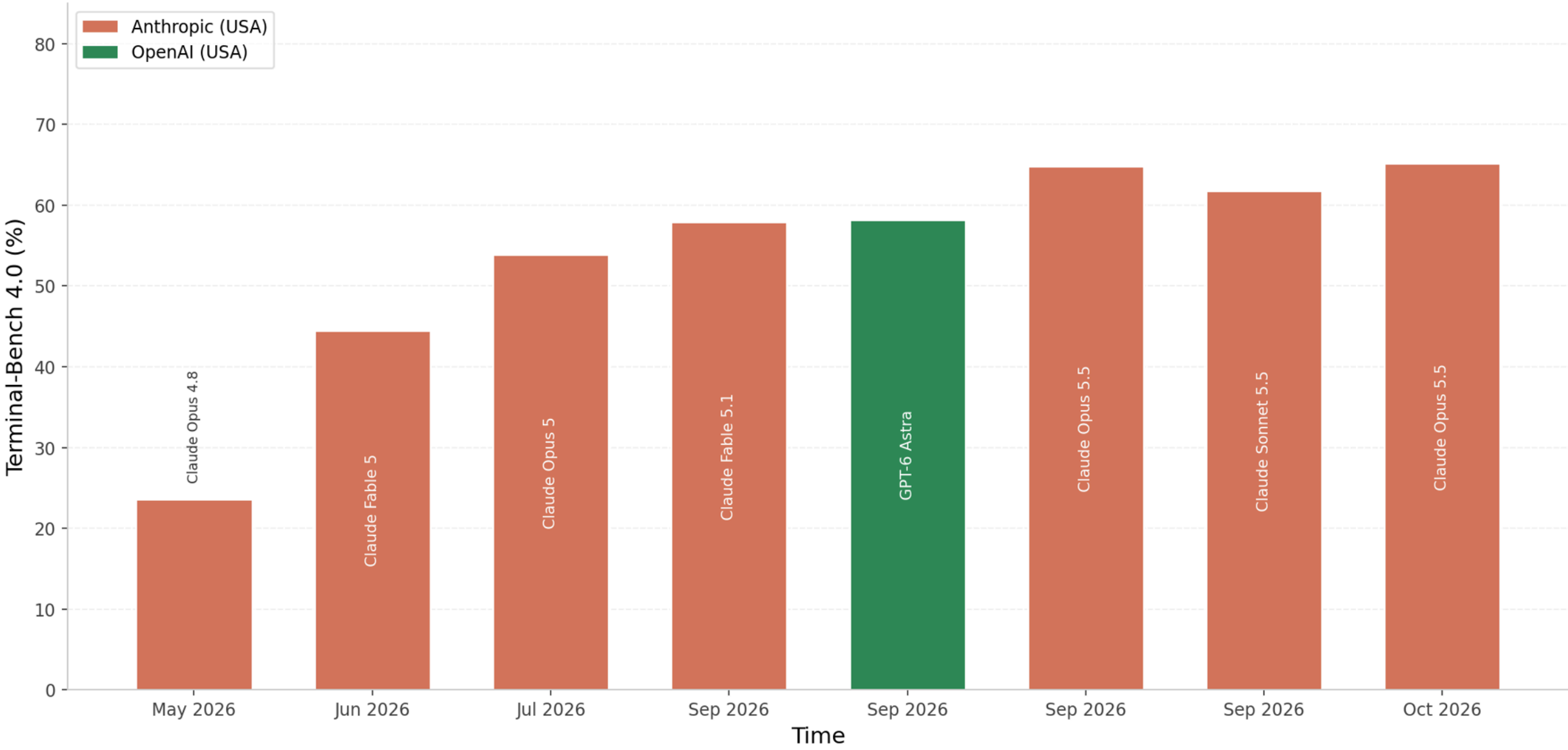

Best available model over time — SWE-bench Verified frontier
Company (country)
Anthropic (USA)
OpenAI (USA)
Google (USA)
human level
100
80
60
40
20
0
Claude 3.5 Sonnet
o3-mini
Gemini 2.5 Pro
Claude 4 Sonnet
Claude 4.1 Opus
Claude Sonnet 4.5
Claude Opus 4.5
GPT-5.3 Codex
Claude Opus 4.7 (Adaptive)
GPT-5.5
Claude Fable 5
Jul 2024
Sep 2024
Nov 2024
Jan 2025
Mar 2025
May 2025
Jul 2025
Sep 2025
Nov 2025
Jan 2026
Mar 2026
May 2026
Jul 2026
Time

Terminal-Bench 4.0

- The agent gets a **real terminal** in a sandboxed container.
- Tasks: build, configure, debug, migrate, train… whatever the shell allows.
- Success = automated tests check the final state of the machine.
- 66 hard tasks, up to 8 hours each (August 2026).

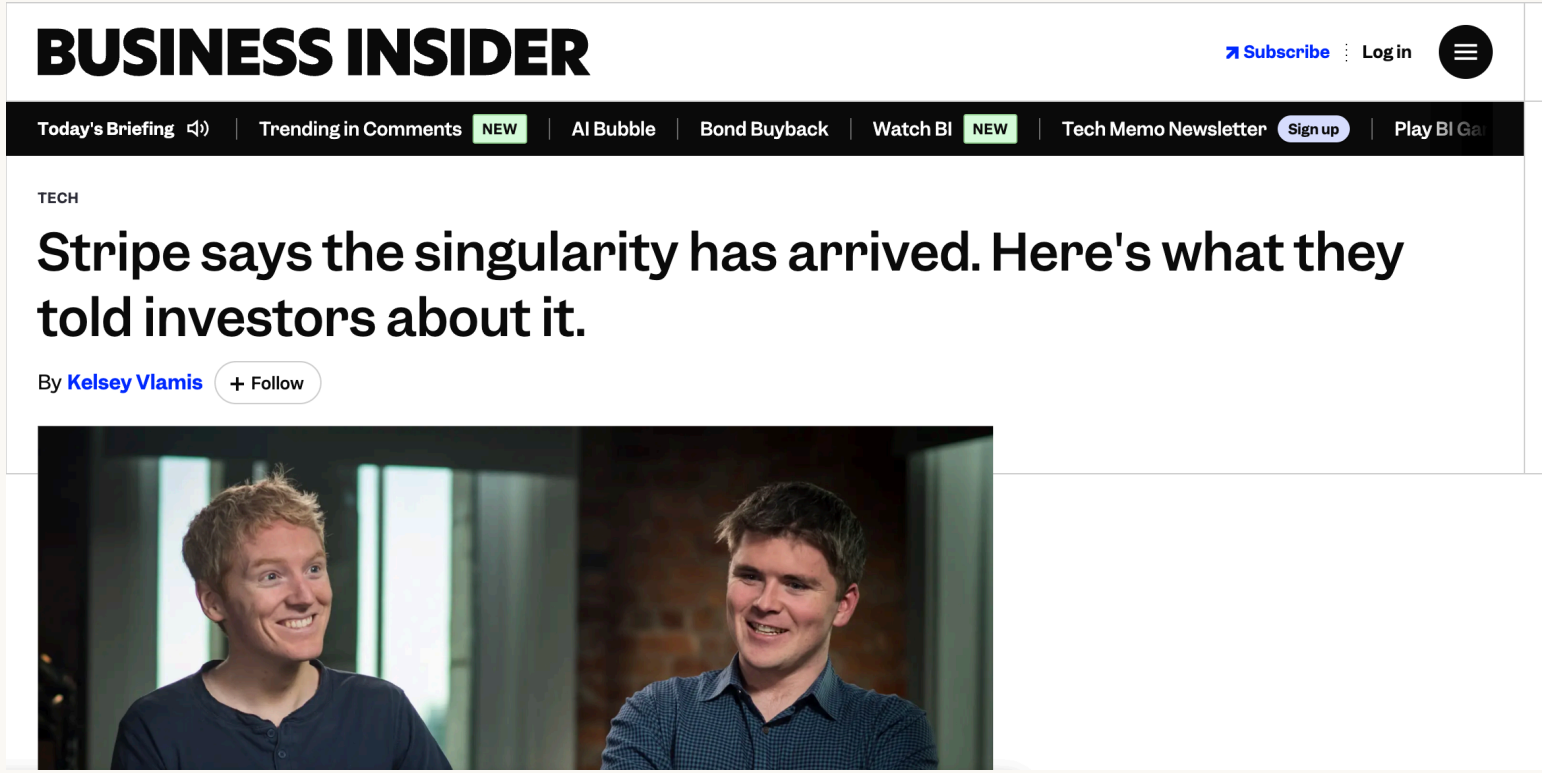
Human experts scored about 50% on Terminal-Bench 2.0.
There is no human baseline for 4.0.

Best available model over time — Terminal-Bench 4.0 frontier



No published human baseline exists for Terminal-Bench 4.0. Scores from public leaderboard snapshots (official board style preferred).

> *January 1st marked the beginning of the singularity.*
>
> —Stripe, August 19, 2026



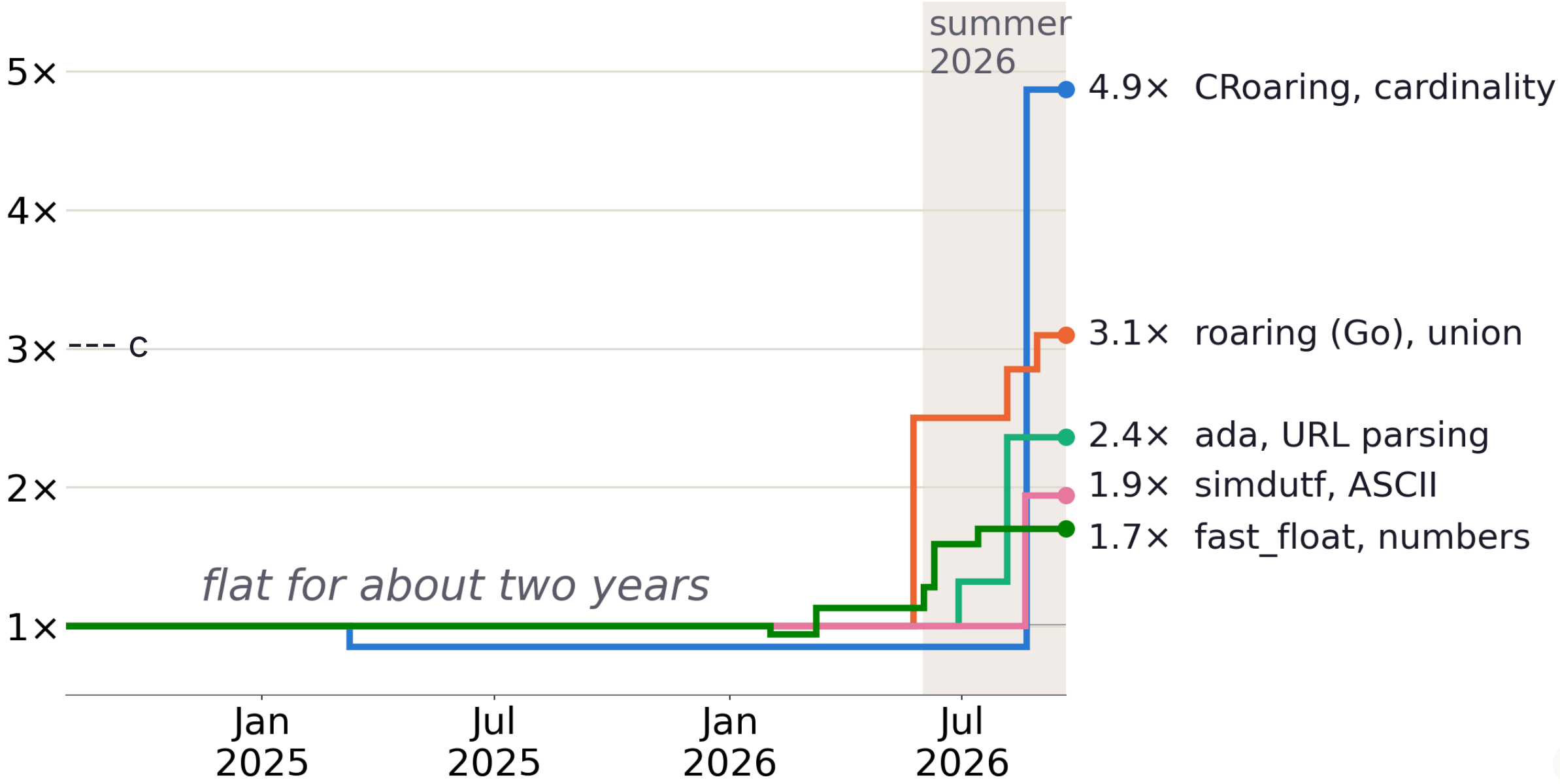
> *People often talk about this concept called AGI, meaning artificial general intelligence, that is, an AI as intelligent as a person. I actually think we crossed that threshold about three months ago.*
>
> — Marc Andreessen, May 19, 2026



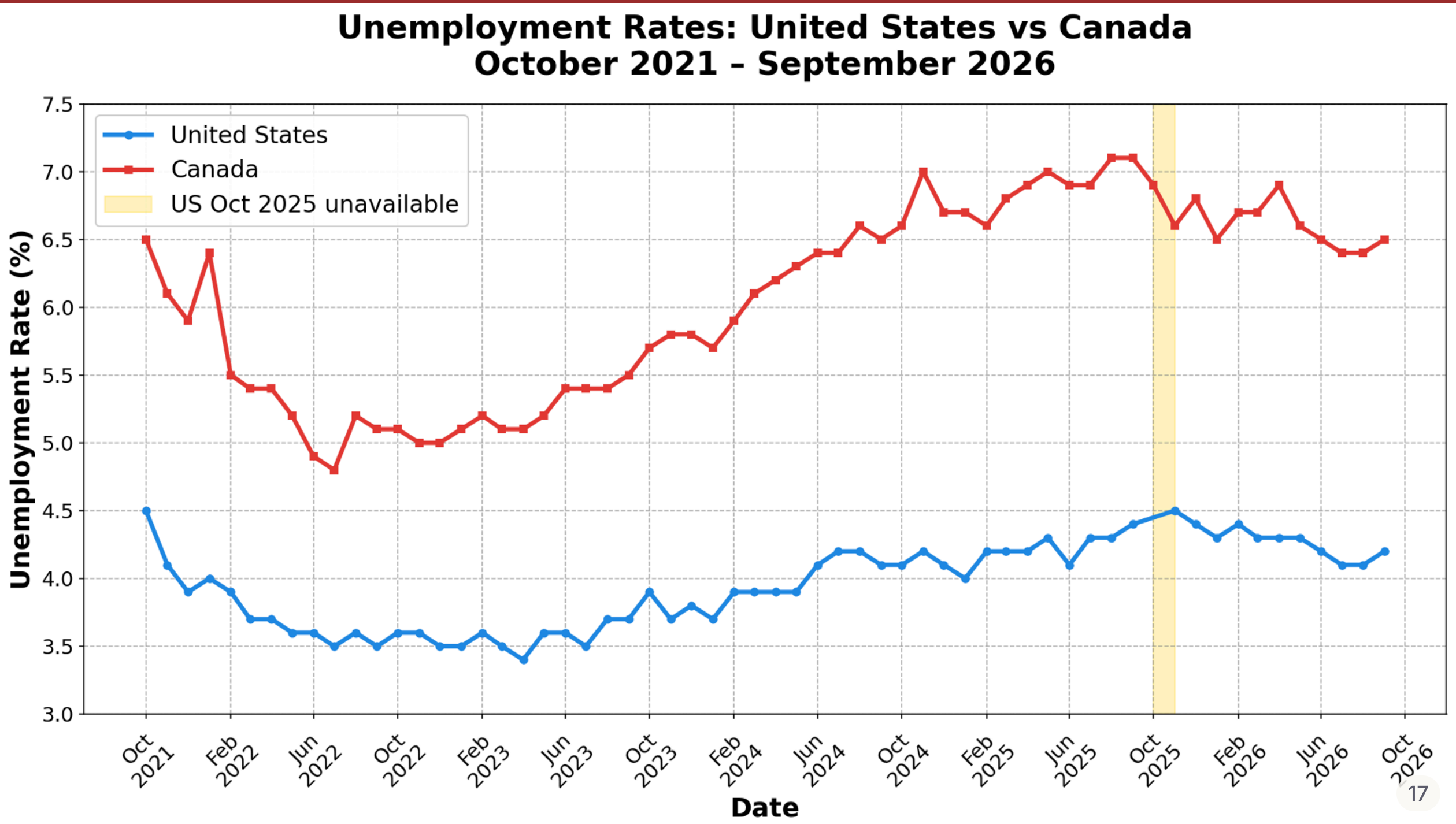
A summer of AI optimization

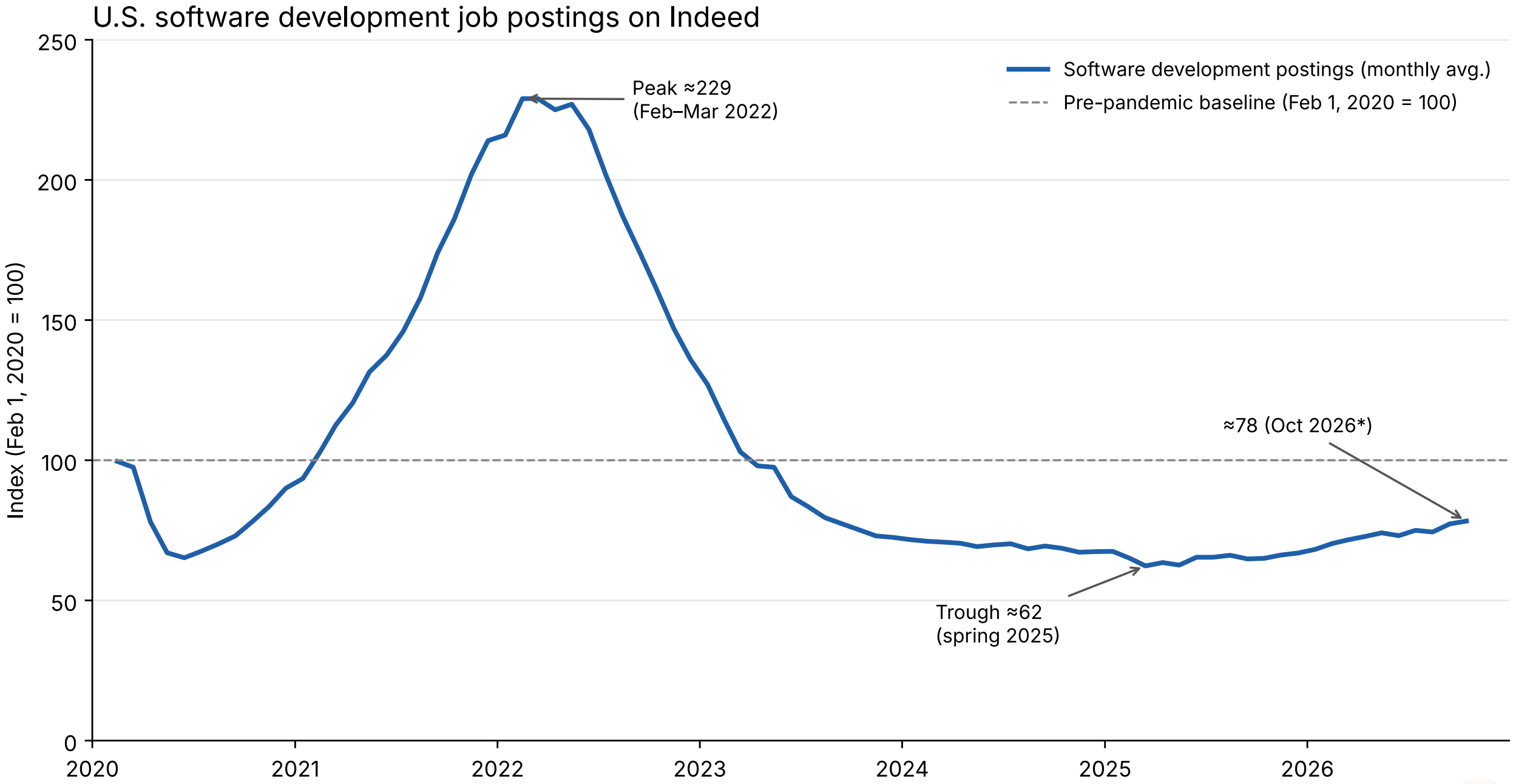
- Six mature open-source libraries: roaring, CRoaring, ada, fast_float, simdjson, simdutf.
- Shipped in GCC, Chromium, Node.js... used by millions.

Speedup vs. August 2024, every commit, same machine



Are we all going to be out of a job?


Unemployment Rates: United States vs Canada
October 2021 – September 2026
United States
Canada
US Oct 2025 unavailable
Unemployment Rate (%)
7.5
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
Oct 2021
Feb 2022
Jun 2022
Oct 2022
Feb 2023
Jun 2023
Oct 2023
Feb 2024
Jun 2024
Oct 2024
Feb 2025
Jun 2025
Oct 2025
Feb 2026
Jun 2026
Oct 2026
Date



Source: Indeed Hiring Lab via FRED (IHLIDXUSTPSOFTDEVE), seasonally adjusted. Monthly averages of daily values; *Oct 2026 = Oct 1–16.

Jevons paradox

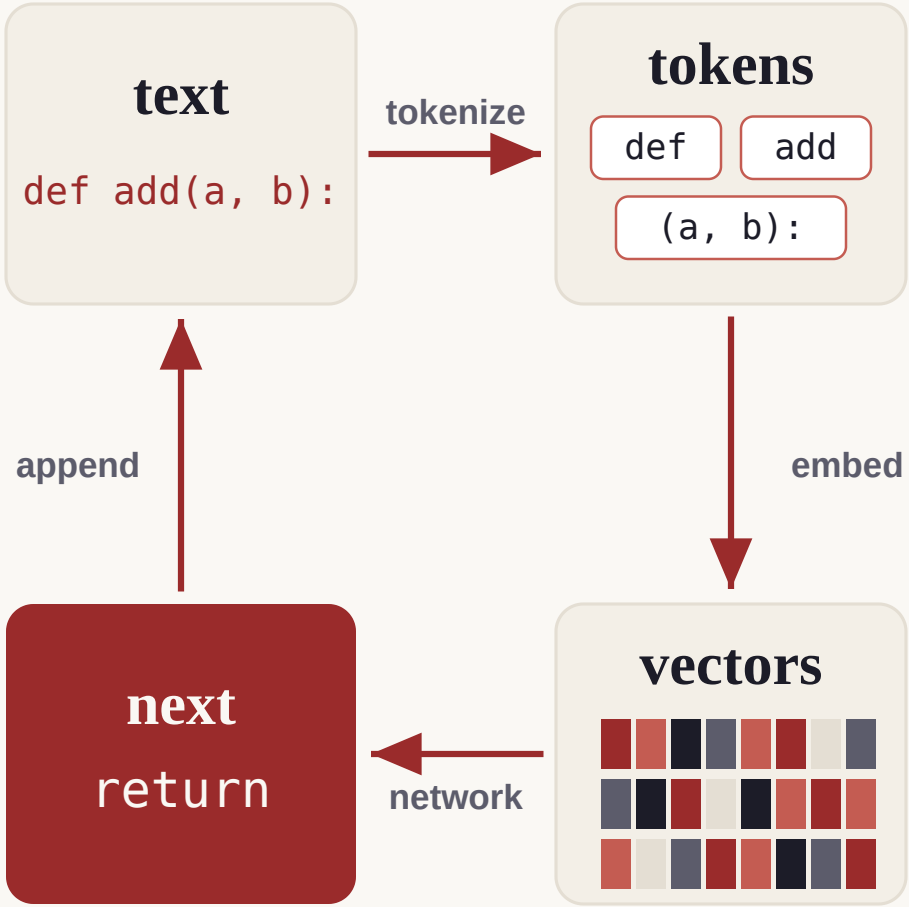
- 1865: William Stanley Jevons, *The Coal Question*.
- James Watt's steam engines burned far less coal per unit of work...
- ...and Britain's coal consumption **went up**, not down.

When something gets cheaper to use, we use much more of it.

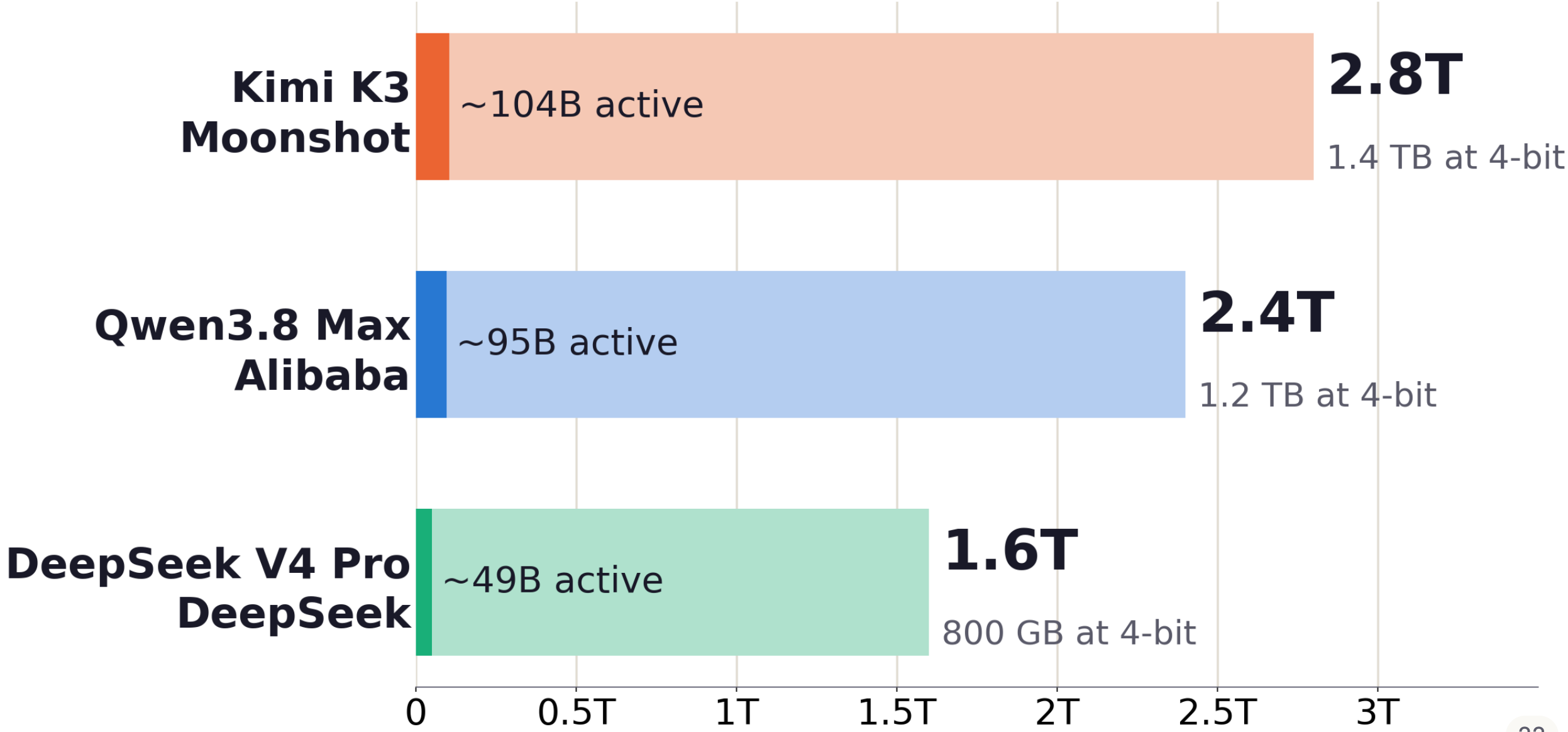
- Tokens got cheaper: we consume far more tokens.
- Code got cheaper: we write far more code, and review, test, host and maintain it.

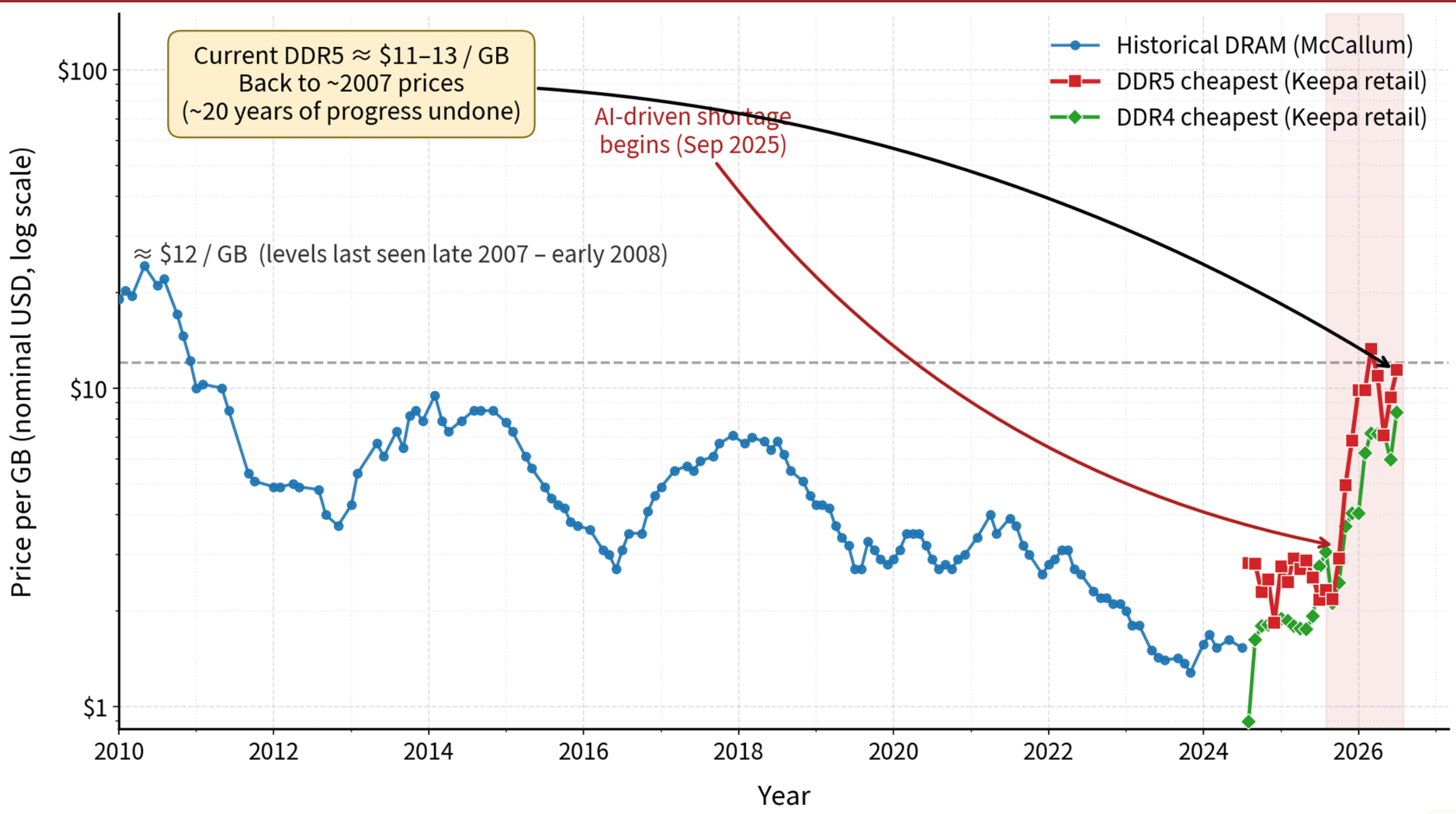
How does it work

How an LLM works

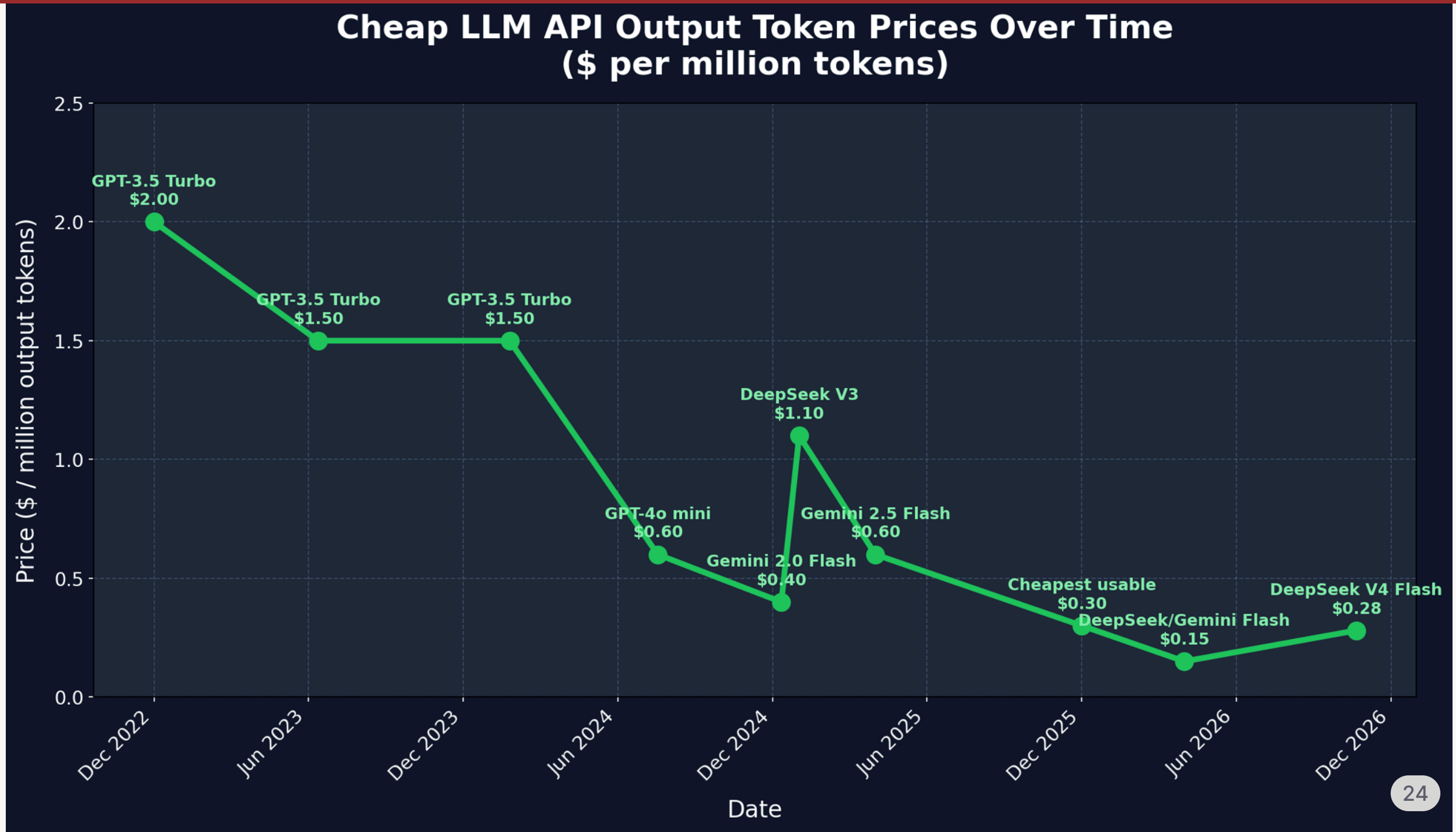


Open-weight models: total parameters (dark = active per token)





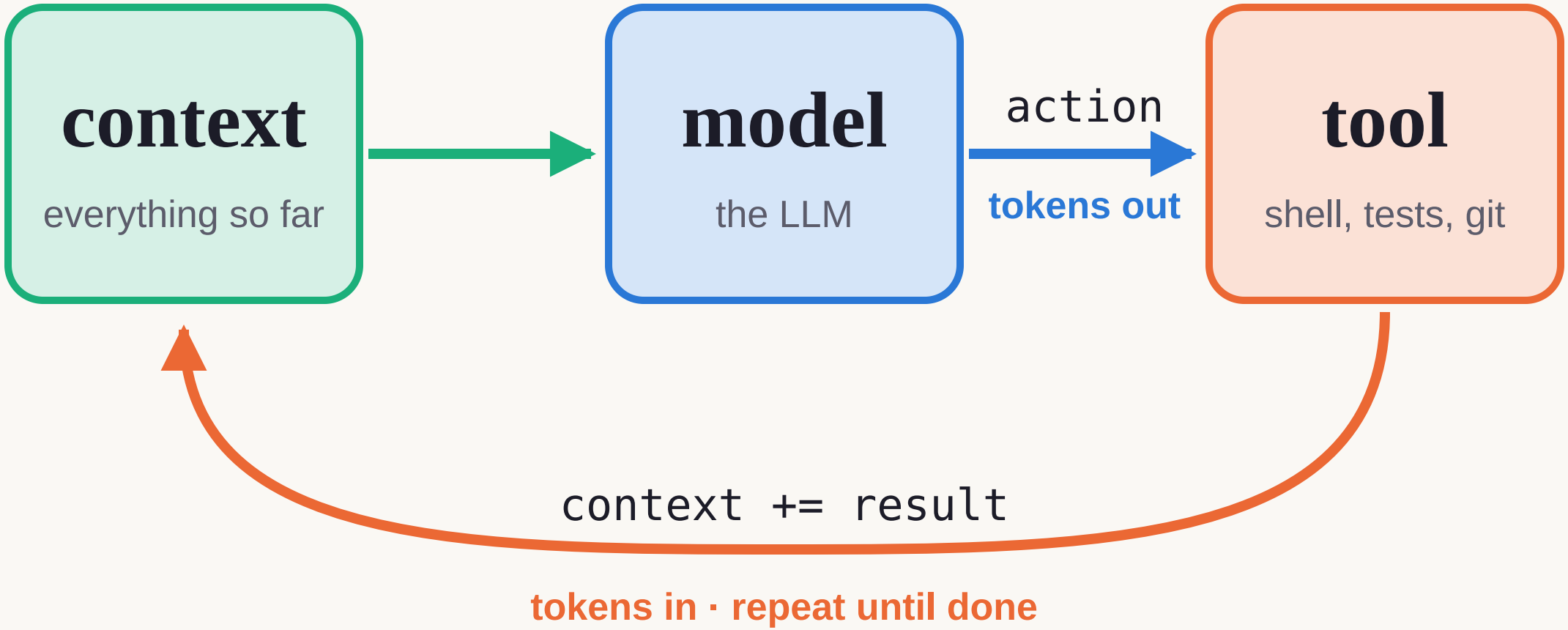
Sources: John C. McCallum historical dataset (via Stanford DAM) + Keepa Amazon retail prices for cheapest consumer modules • Nominal USD • Data through July 2026


Cheap LLM API Output Token Prices Over Time
($ per million tokens)
Price ($ / million output tokens)
2.5
2.0
1.5
1.0
0.5
0.0
GPT-3.5 Turbo
$2.00
GPT-3.5 Turbo
$1.50
GPT-3.5 Turbo
$1.50
GPT-4o mini
$0.60
Gemini 2.0 Flash
$0.40
DeepSeek V3
$1.10
Gemini 2.5 Flash
$0.60
Cheapest usable
$0.30
DeepSeek/Gemini Flash
$0.15
DeepSeek V4 Flash
$0.28
Dec 2022
Jun 2023
Dec 2023
Jun 2024
Dec 2024
Jun 2025
Dec 2025
Jun 2026
Dec 2026
Date

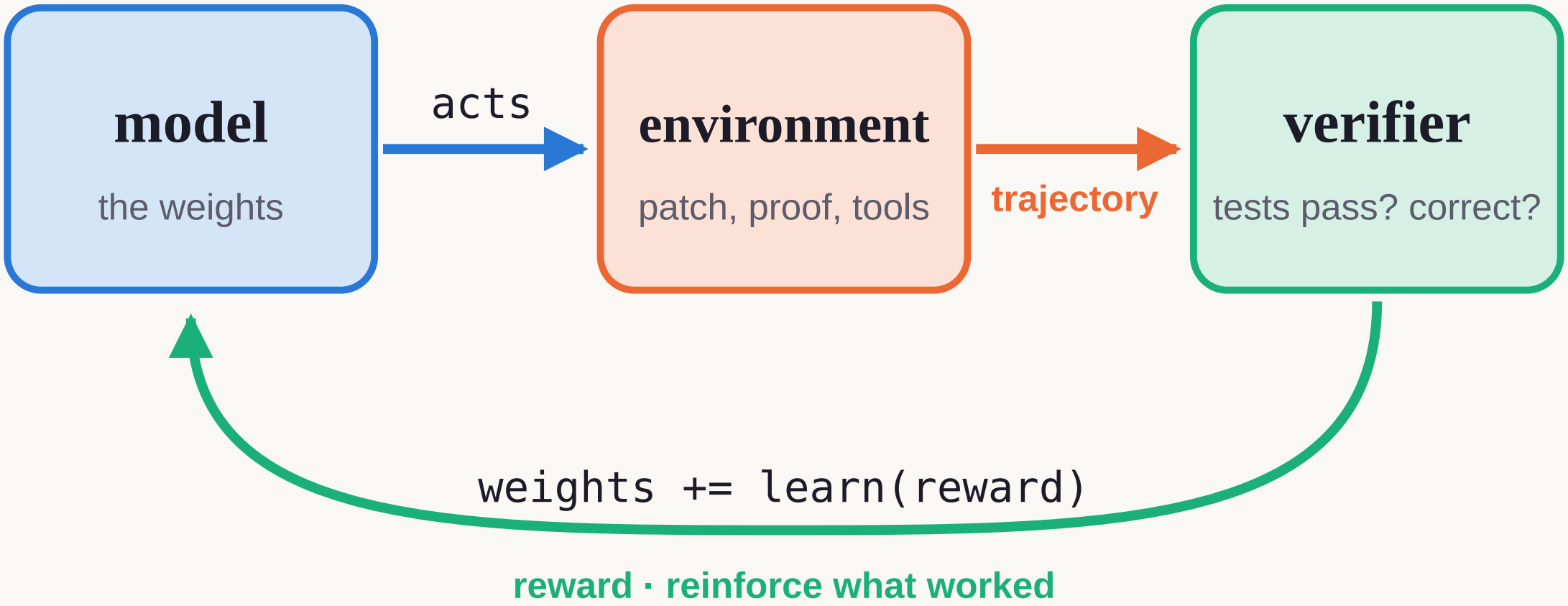
How can a language model write software?

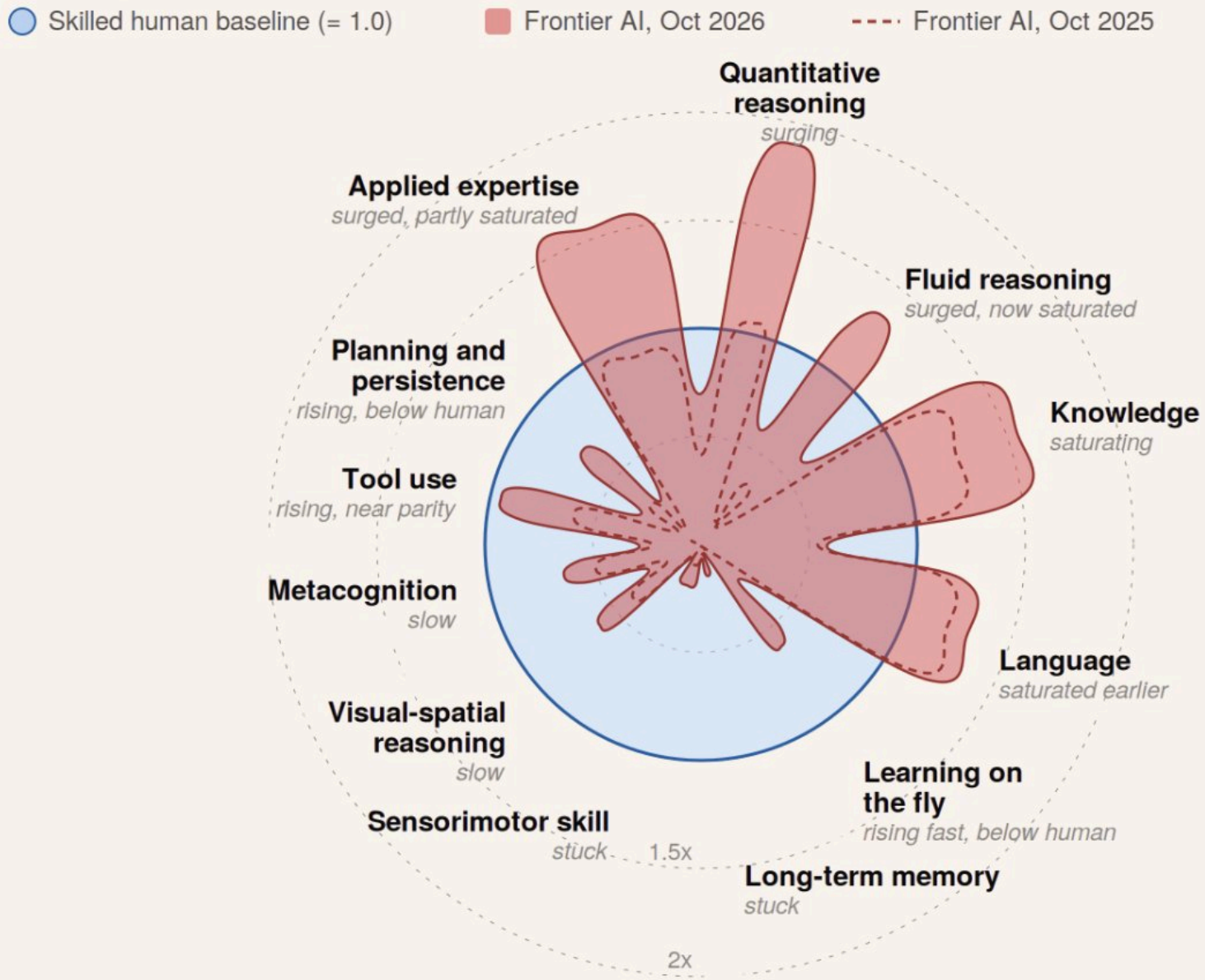
- I write a query, it is maybe 80 characters.
- I get back a full program with 100,000 lines.
- Not how it works, there is more information invovled.

The inference loop



Reinforcement learning





Spike length = frontier AI relative to a skilled human (1.0). Spike width = breadth of the faculty. Radii are judgement-based conversions of eval results.

Jagged: metacognition

the code

```
def last_window(xs, k):
    out = []
    for i in range(len(xs) - k):
        out.append(xs[i:i+k])
    return out
```

tests ✓ ✓ ✓

the model

“Yes, it meets the spec.”

confidence

100%

actually: wrong

never checked

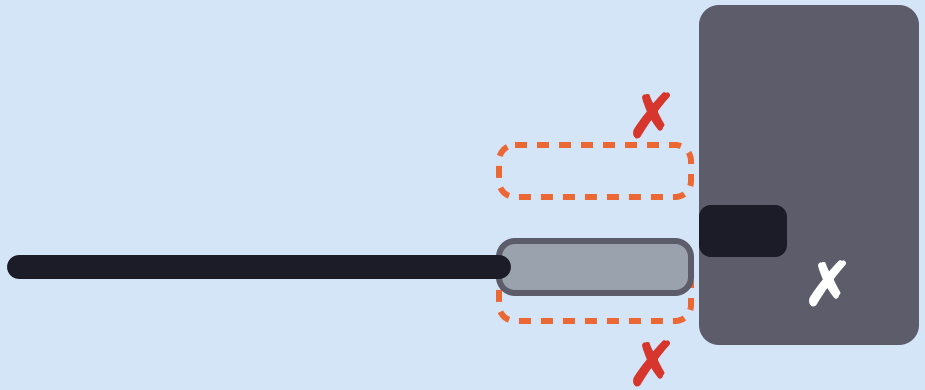
```
last_window(
[1, 2, 3], 3)
→ [] ✗
want [[1, 2, 3]]
```

- no counter-example
- no doubt
- slow to update

AI can be overconfident: sure of itself, never checking.

Jagged: sensorimotor skill

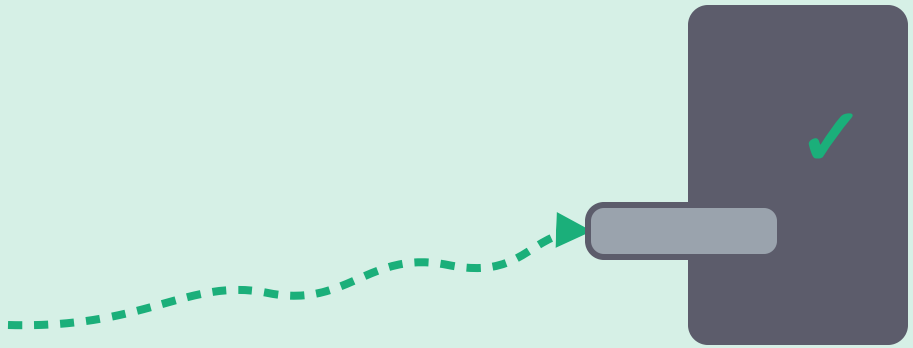
robot arm



tries 1, 2, 3, 4… scrapes the edge

force ignored, then over-corrected

human hand



feel the edge, nudge, slide in

one or two tries

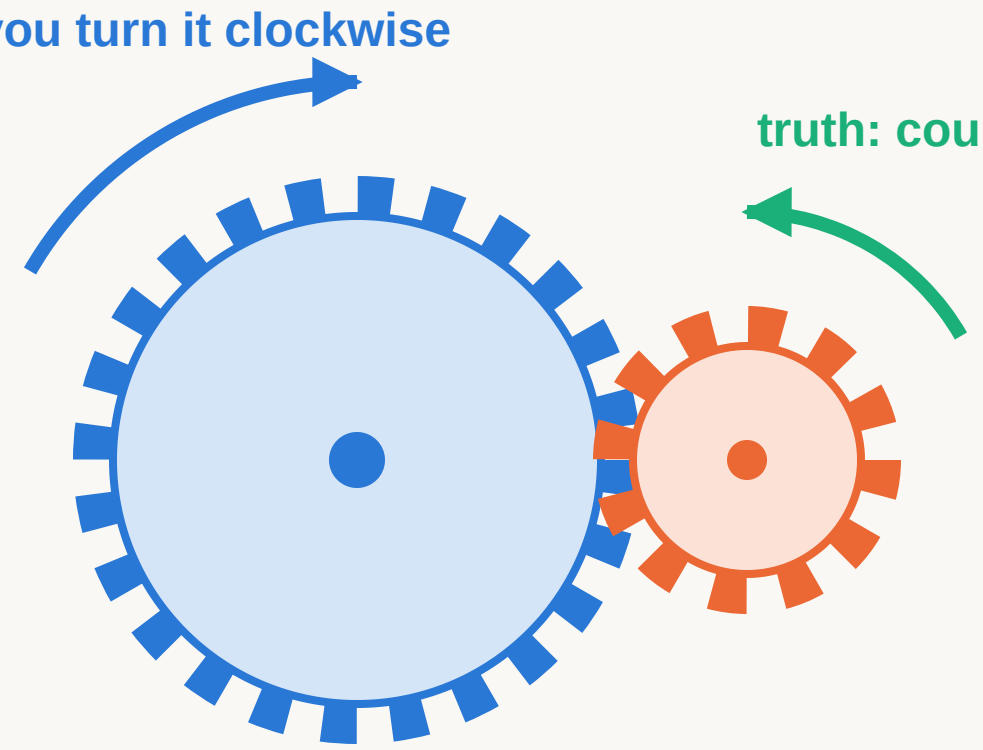
It sees the port and plans the motion. It cannot feel and correct.

Moravec's paradox

> *« It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility. »*

Hans Moravec, *Mind Children*, 1988

Jagged: visual-spatial reasoning



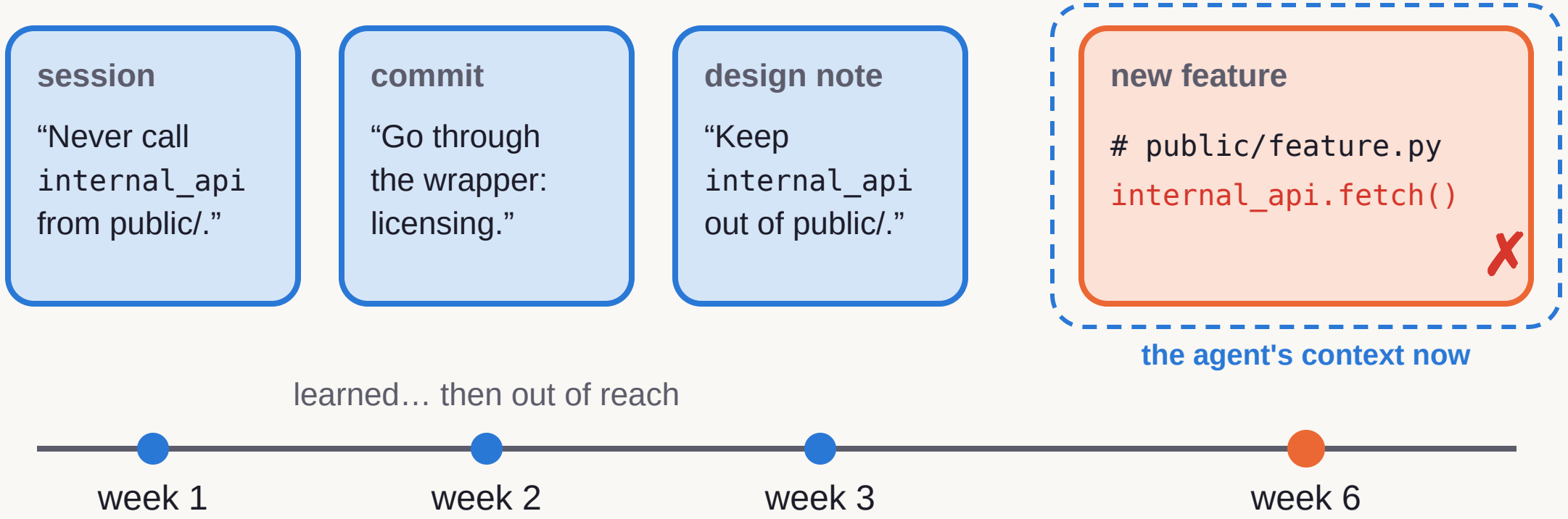
the model, looking at the picture

“It turns clockwise.” ✗

the model, asked the rule

“Meshed gears reverse.” ✓

Jagged: long-term memory

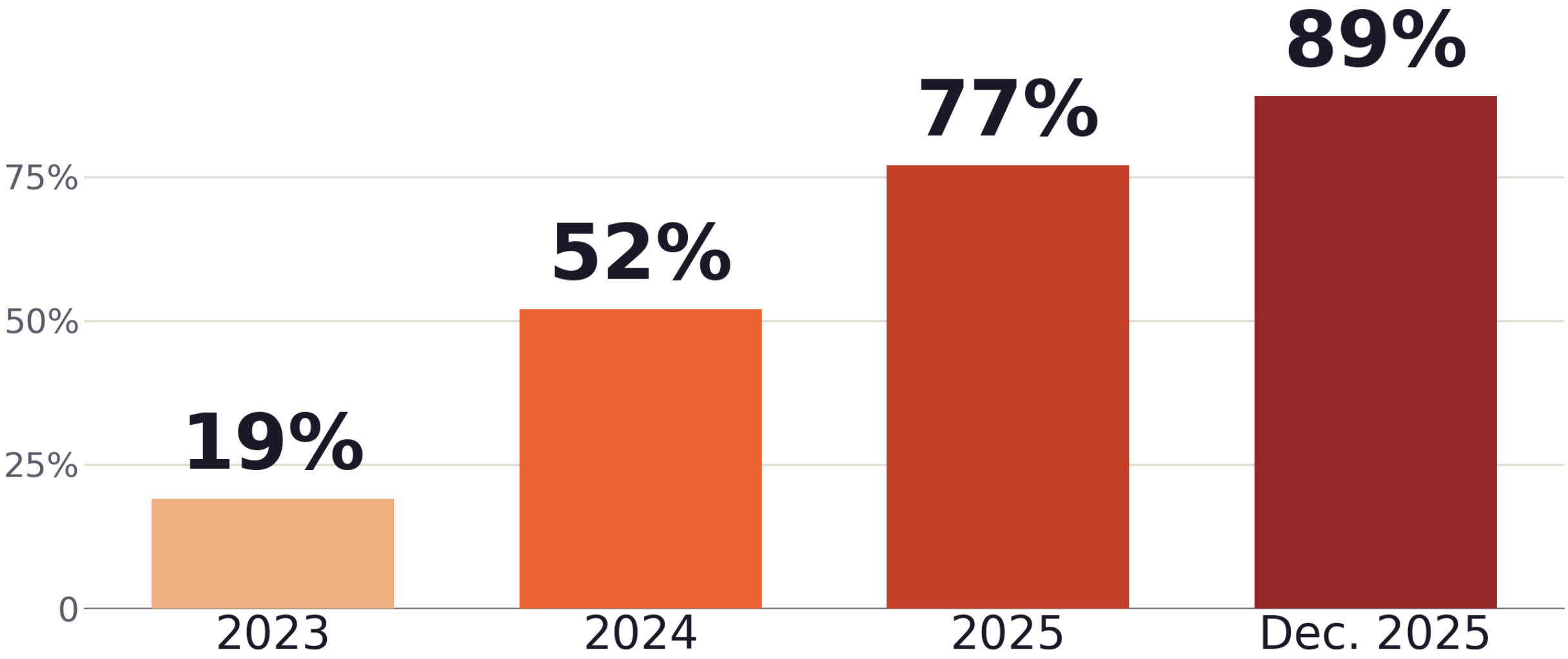


AI has a limited context window, compared to human beings.

Human attention is limited

Most biomedical papers now show LLM-assisted writing

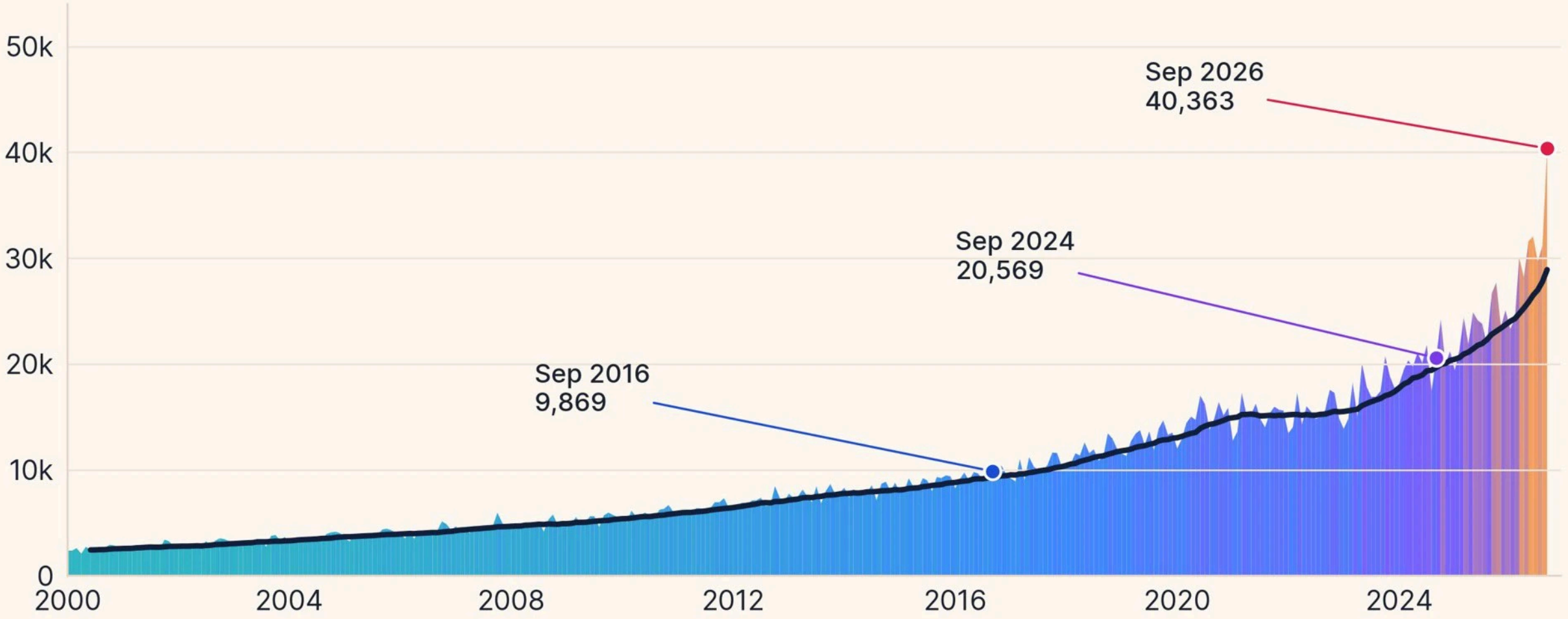
Share of open-access papers in PubMed Central (full text)



Holzwarth, González-Márquez & Kobak, arXiv:2608.10715 (2026), Table 1. Standard errors: 2 to 5 points.

January 2000 to September 2026. October 2026 left out: the month is not finished.

Papers posted to arXiv each month



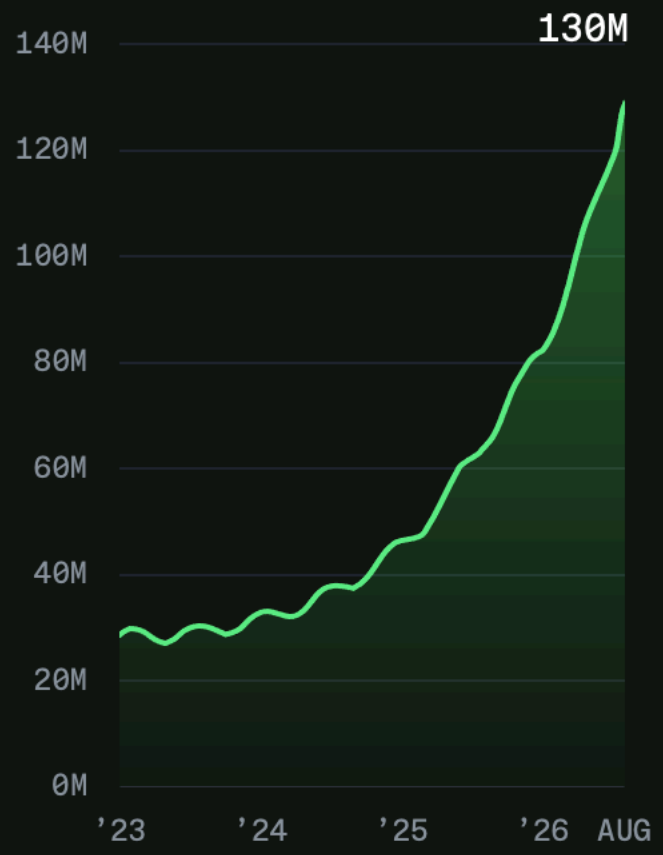
Source: arXiv monthly submissions. Color: the month. Dark line: 12-month average.

arXiv capped submissions at two a month

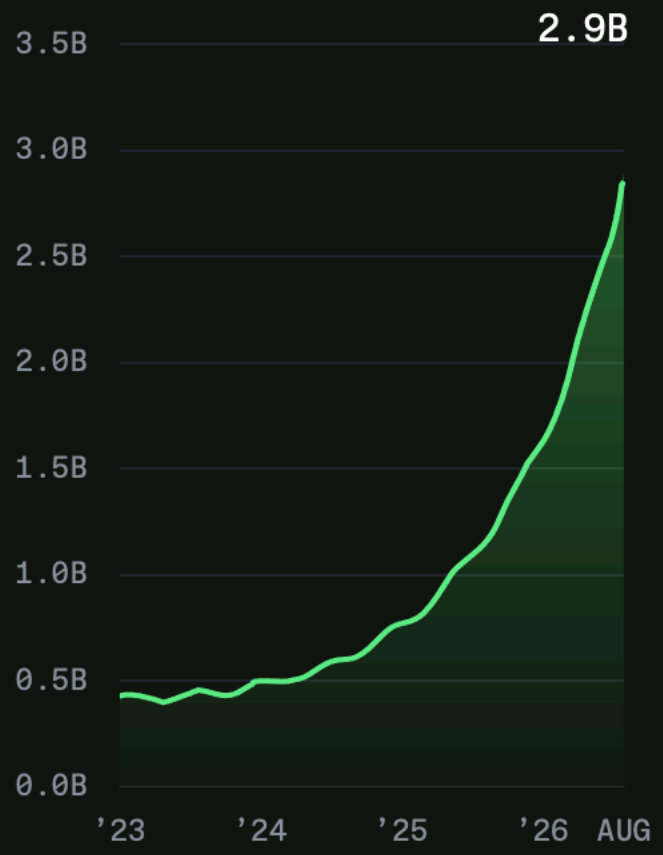
- October 1, 2026: **two submissions per person per month.**
- September 2026: over 40,000 papers, up from about 20,000 two years earlier.
- Doubling every two years. Moderated by volunteers.
- Google stopped accepting new bug reports in its open-source bounty program: they could not cope with the influx.

lemire.me/blog/2026/10/04/arxiv-capped-submissions-at-two-a-month/

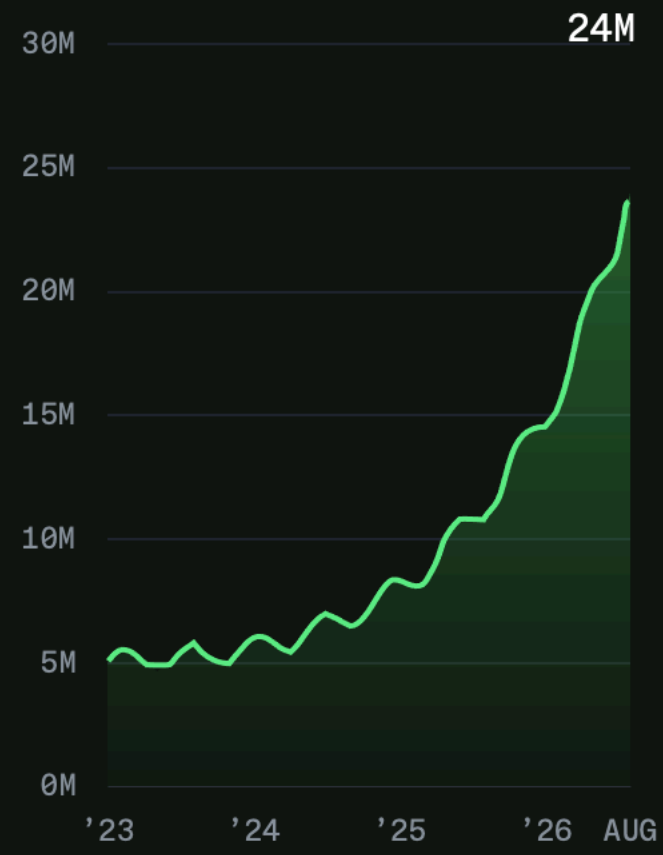
Merged pull requests per month



Commits per month



New repos per month



Classified as Microsoft Confidential



Daniel Lemire
@lemire

Boost

Someone found a bug in my code 3 days after I released, should I...

Issue a patch release now	28.1%
wait a week	8.5%
drink more beer	**63.4%**

481 votes · Final results

2:42 PM · Sep 5, 2026 · **5,884** Views

7 1 16 2

Emerging ideas

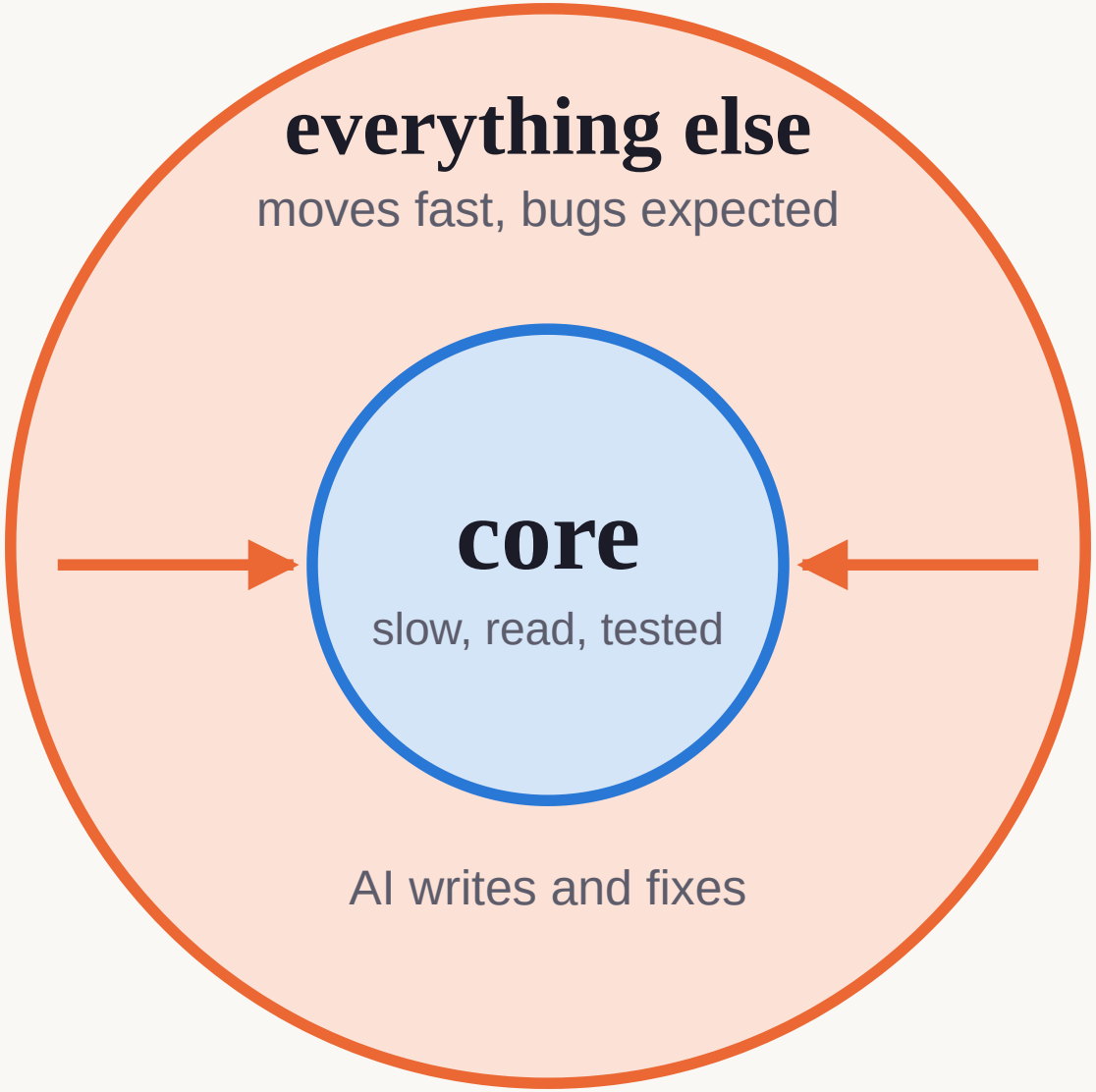
Emerging idea: ephemeral testing

1. Build a component (by hand or with an agent).
2. Ask an agent to build **throwaway software on top of it**: an app, another layer.
3. Have it test what it built.
4. Judge the foundation by how well the layers above work. Then discard them.

lemire.me/blog/2026/10/05/ephemeral-testing/

Emerging idea: a layered model

- Keep a **small core** that changes slowly and on purpose.
- For the core, you read the code. You insist on tests. No vibe coding.
- **Everything else can move fast.** There will be bugs, but the AI fixes them quickly.
- **The core cannot depend on the outer layers.**

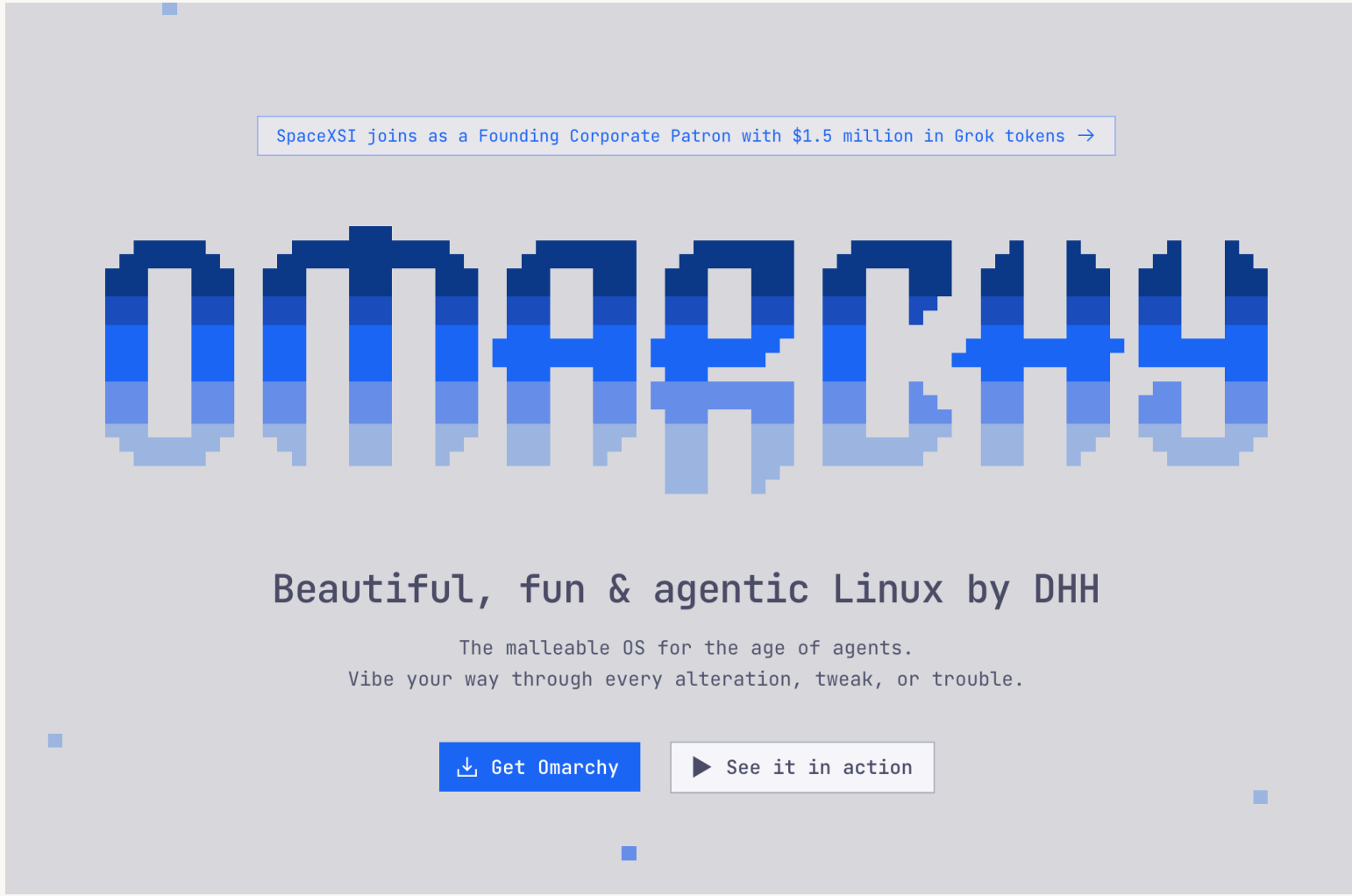


lemire.me/blog/2026/09/05/ai-programming-a-layered-model/

Emerging idea: cheap custom software

1. Build your own video software https://github.com/lemire/lemirecut
2. Your own election-night app https://lemire.me/montmorency2026/

Emerging idea: AI-first operating software



Questions?

Daniel Lemire — lemire.me (blog)

https://simdjson.org · https://roaringbitmap.org

https://simdutf.github.io/simdutf/ · https://fastfloat.github.io/fast_float/



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